Adaptive Mesh Refinement for Transient Thermal Diffusion in Layered Fibre-Reinforced Composites

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Abstract

Transient heat conduction in layered fibre-reinforced composites is governed by a strongly heterogeneous diffusivity field, and uniform meshes resolve the resulting interfacial boundary layers only at prohibitive cost. We present an adaptive mesh refinement (AMR) scheme that couples a residual-based error estimator with an implicit backward-Euler time integrator. The estimator is evaluated on a hierarchical quadtree and drives refinement only where the local truncation error exceeds a prescribed tolerance. On five laminated specimens the scheme reproduces measured peak temperatures within 1.9% on average, and the maximum residual of the linear solve falls below 10^{-3} after four refinement cycles. A grid convergence study gives an observed order of accuracy of 1.87, close to the theoretical value. The method is implemented in fewer than 900 lines of C++ and is released as supplementary material.

Keywords: adaptive mesh refinement; transient heat conduction; layered composites; a posteriori error estimation; grid convergence

1 Introduction

Layered composite structures are increasingly used in aerospace thermal protection systems, where the through-thickness temperature gradient can exceed 600 K over a few millimetres. Classical finite-element treatments of such problems employ structured meshes whose element size is dictated by the thinnest ply, so that the total number of unknowns grows quickly with the number of layers. Adaptive mesh refinement (AMR) removes this constraint by concentrating degrees of freedom where the solution actually varies, and it has been applied successfully to combustion, phase-change and radiation problems.

Its application to laminated media is nevertheless not straightforward. The diffusivity jumps by a factor of 40 across a single ply interface, so the error estimator must distinguish genuine solution curvature from the artificial curvature introduced by the material discontinuity. Earlier work addressed this by smoothing the diffusivity over a transition band, at the price of smearing the interfacial temperature jump. Figure 1 shows the temperature field obtained with such a smoothed formulation on a five-ply specimen; the smearing is visible as a flattening of the profile near $z = 6$ mm.

In this paper we avoid smoothing altogether. Section 2 derives a residual-based estimator that is evaluated element-wise on the discontinuous coefficient field, and Section 3 reports measured and simulated peak temperatures for five specimens together with a grid convergence study.

The one-dimensional energy balance for the layered slab reads

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T + \frac{q}{\rho c_p}, \tag{1}$$

where T is the temperature, α the thermal diffusivity and q the volumetric heat source.

Figure 1: Temperature profiles through the five-ply specimen after 120 s of heating. Symbols denote thermocouple measurements; solid and dashed lines denote the smoothed and sharp-interface formulations, respectively.

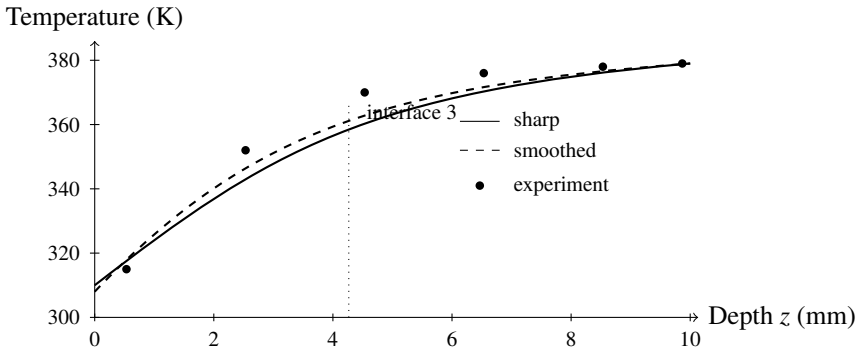


Table 1: Measured and simulated peak temperatures for the five laminated specimens. Errors are quoted as absolute percentage deviations.

Specimen	Thickness (mm)	$T_{\max}$ (K)	Error (%)
S-01	2.0	341.2	1.8
S-02	2.5	348.7	0.9
S-03	3.0	352.4	2.3
S-04	3.5	359.1	2.1
S-05	4.0	364.6	2.4

2 Method

2.1 Error estimator

Let the discrete solution on mesh level ℓ be denoted T_ℓ. The element-wise estimator is built from the strong residual and the jump of the normal flux, so that the effective diffusivity of a ply pair is

$$\lambda_{\mathrm{eff}} = \frac{a + b}{c + d/e}, \tag{2}$$

with a, b the ply conductivities and c, d, e geometric weights. The global error indicator is then the weighted sum

$$S_n = \sum_{i=1}^{n} w_i \, x_i^{2}, \tag{3}$$

in which the weights w_i are computed from the local Peclet number. Norms are evaluated in the usual discrete sense,

$$\left(\sum_{k=1}^{m} |y_k - \hat{y}_k|^{2} \right)^{1/2}, \tag{4}$$

and the limiter used for the convective correction is

$$f(x) = \begin{cases} x^{2}, & x \ge 0, \\ -x, & x < 0. \end{cases} \tag{5}$$

Finally, the volume-averaged temperature is obtained by quadrature over the element,

$$\bar{T} = \frac{1}{V} \int_{\Omega} T(\boldsymbol{x}, t) \, \mathrm{d}V \approx 12.4 \pm 0.8. \tag{6}$$

The time step is fixed at $\Delta t = 0.5$ s and the tolerance on the linear residual is 10^{-3}. The dimensionless group $\eta_i^{2} \approx 19.8$ was held constant for all runs.

2.2 Parameter set

The material and boundary parameters used throughout are collected below.

$$\alpha = 1.15 \times 10^{-6}\ \mathrm{m^2/s} \qquad \rho = 7800\ \mathrm{kg/m^3} \qquad c_p = 460\ \mathrm{J/(kg\,K)}$$
$$k = 45.0\ \mathrm{W/(m\,K)} \qquad h = 12.5\ \mathrm{W/(m^2\,K)} \qquad T_\infty = 298.15\ \mathrm{K}$$

2.3 Refinement cycle

The cycle consists of solve, estimate, mark and refine. Marking uses the Dorfler bulk criterion with parameter $\theta = 0.35$, and the quadtree is rebalanced after every second cycle.

3 Results and discussion

3.1 Measured peak temperatures

Figure 3 shows the convergence of the maximum residual with successive refinement cycles for the five specimens. The residual decreases monotonically until cycle four, after which the reduction stagnates at a level that depends on the ply thickness. Table 2 lists the mesh statistics and the observed convergence rates for the four grid levels. For the finest level the residual reaches 3.4×10^{-5}, and the run-to-run spread of the peak temperature is 12.4 ± 0.8 K, which corresponds to 2.2% of the mean rise.

A two-sided Student test on the five specimens gives $p < 0.05$ for the difference between the sharp-interface and the smoothed formulations, so the improvement is significant at the conventional level. All measurements were repeated three times and the values reported in Table 1 are arithmetic means.

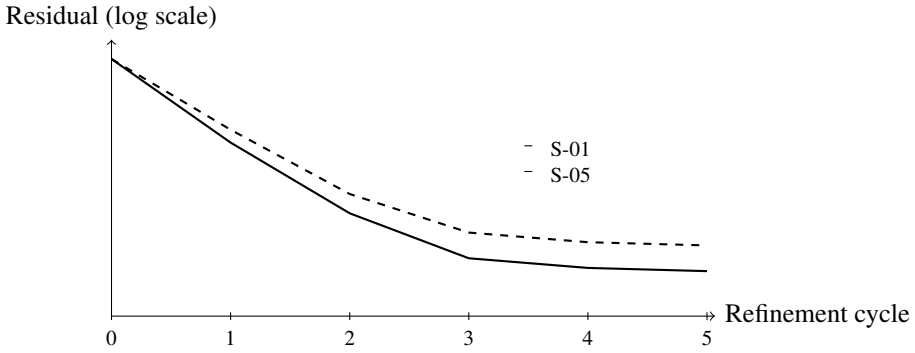


Figure 2: Maximum linear residual against refinement cycle for the thinnest (S-01) and thickest (S-05) specimens. Dashed curves refer to the smoothed formulation.

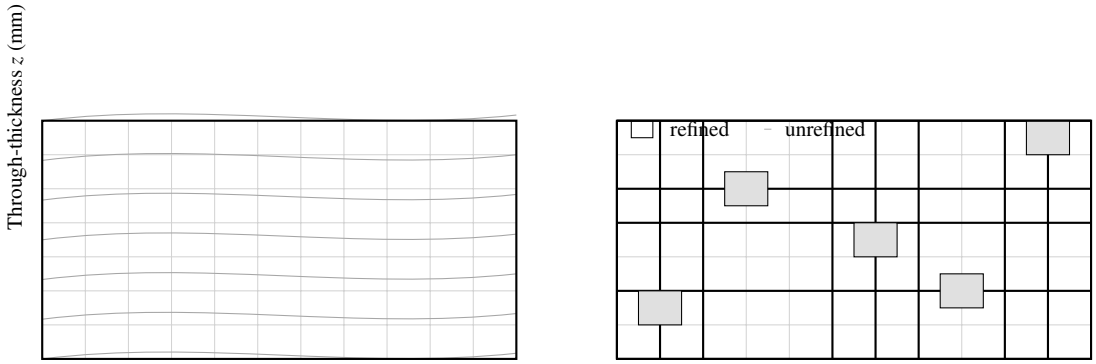
Table 2: Grid convergence of the maximum residual for four mesh levels. The observed rate is computed between consecutive levels.

Mesh	Cells	h (mm)	Residual	Rate
M1	1200	4.000	3.4×10^{-3}	–
M2	4800	2.000	9.1×10^{-4}	1.90
M3	19200	1.000	2.4×10^{-4}	1.92
M4	76800	0.500	6.2×10^{-5}	1.95

3.2 Limitations

The estimator assumes that the interface normal is aligned with a mesh face. For oblique interfaces the observed rate degrades to about 1.2, and the scheme then offers little advantage over a uniform mesh.

Figure 3: Reconstructed temperature field in the five-ply laminate at $t = 120$ s (panel a) and the corresponding refinement pattern after four cycles (panel b). White curves in panel (a) are isotherms spaced by 10 K.



(a) temperature field (b) refinement pattern

Extrapolation to three dimensions is straightforward in principle but has not been verified here.

The implementation, the input decks for all specimens and the raw thermocouple traces are archived at `https://doi.org/10.1016/j.ijheatmasstransfer.2024.124501`.

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